

Analysis of Trader Performance Based on Bitcoin Market Sentiment

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Company: Primetrade.ai

Tool Used: Google Colab, Python, Pandas, Matplotlib

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1. Introduction

This project analyzes the relationship between Bitcoin market sentiment and trader performance using two datasets: Bitcoin Fear & Greed sentiment data and historical trader data from Hyperliquid. The main objective of this analysis is to understand how trader profitability, win rate, trade side, leverage usage, symbols, and account-level performance vary under different market sentiment conditions.

Market sentiment plays an important role in trading behavior. During Fear conditions, traders may act more carefully, while during Greed conditions, traders may take more aggressive positions. This analysis helps identify hidden patterns in trader performance and provides insights that can support smarter trading strategies.

2. Dataset Description

Two datasets were used for this analysis.

2.1 Bitcoin Market Sentiment Dataset

The Bitcoin Market Sentiment dataset contains date-wise sentiment classification. The main columns used from this dataset are:

- Date
- Classification / Sentiment

The sentiment classification includes market conditions such as Fear and Greed.

2.2 Historical Trader Data

The Historical Trader Data from Hyperliquid contains detailed trading information. The important columns used for analysis include:

- Account
- Symbol
- Execution Price
- Size
- Side
- Time
- Closed PnL
- Leverage

The Closed PnL column was used to measure trader profit or loss.

3. Tools and Technologies Used

The following tools and technologies were used in this project:

- Google Colab

- Python
- Pandas
- NumPy
- Matplotlib
- Microsoft Excel
- Microsoft Word

Google Colab was used for data cleaning, analysis, and visualization. Excel was used to store analysis results, and Word was used to prepare the final report.

4. Methodology

First, both datasets were uploaded into Google Colab. The column names were cleaned to make them easy to use in Python. The date column from the sentiment dataset and the time column from the trader dataset were converted into a common date format.

After cleaning the dates, both datasets were merged using the date column. This helped connect each trade with the market sentiment on that particular date.

Trades with zero Closed PnL were removed from the final performance analysis because they do not represent actual profit or loss. After this, the final analysis was performed using the following metrics:

- Total PnL
 - Average PnL
 - Win Rate
 - Total Number of Trades
 - Side-wise Performance
 - Symbol-wise Performance
 - Account-wise Performance
 - Leverage Analysis
 - Daily PnL Trend
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5. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

1. Loaded both CSV datasets into Google Colab.
2. Checked dataset shape, column names, and missing values.

3. Cleaned column names by converting them into lowercase and removing extra spaces.
4. Converted date and time columns into proper date format.
5. Converted Closed PnL values into numeric format.
6. Merged trader data with sentiment data using the date column.
7. Removed zero PnL trades for accurate performance analysis.

After preprocessing, the cleaned dataset was used for analysis.

6. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand how trader performance changes under different Bitcoin market sentiment conditions.

The following analyses were performed:

- Sentiment-wise PnL analysis
 - Average PnL by sentiment
 - Total PnL by sentiment
 - Win rate by sentiment
 - Buy/Sell side analysis
 - Symbol-wise performance analysis
 - Account-wise performance analysis
 - Leverage impact analysis
 - Daily PnL trend analysis
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7. Analysis and Results

7.1 Sentiment-wise PnL Analysis

The sentiment-wise PnL analysis was performed to compare trader profitability across different market sentiment conditions. The analysis showed that trader performance changes depending on whether the market sentiment is Fear or Greed.

Insert Chart Here: Average PnL by Market Sentiment

Caption: Figure 1: Average Trader PnL by Market Sentiment

From the analysis, the highest average PnL was observed during Greed sentiment. This means that, on average, individual profitable trades performed better during Greed market conditions.

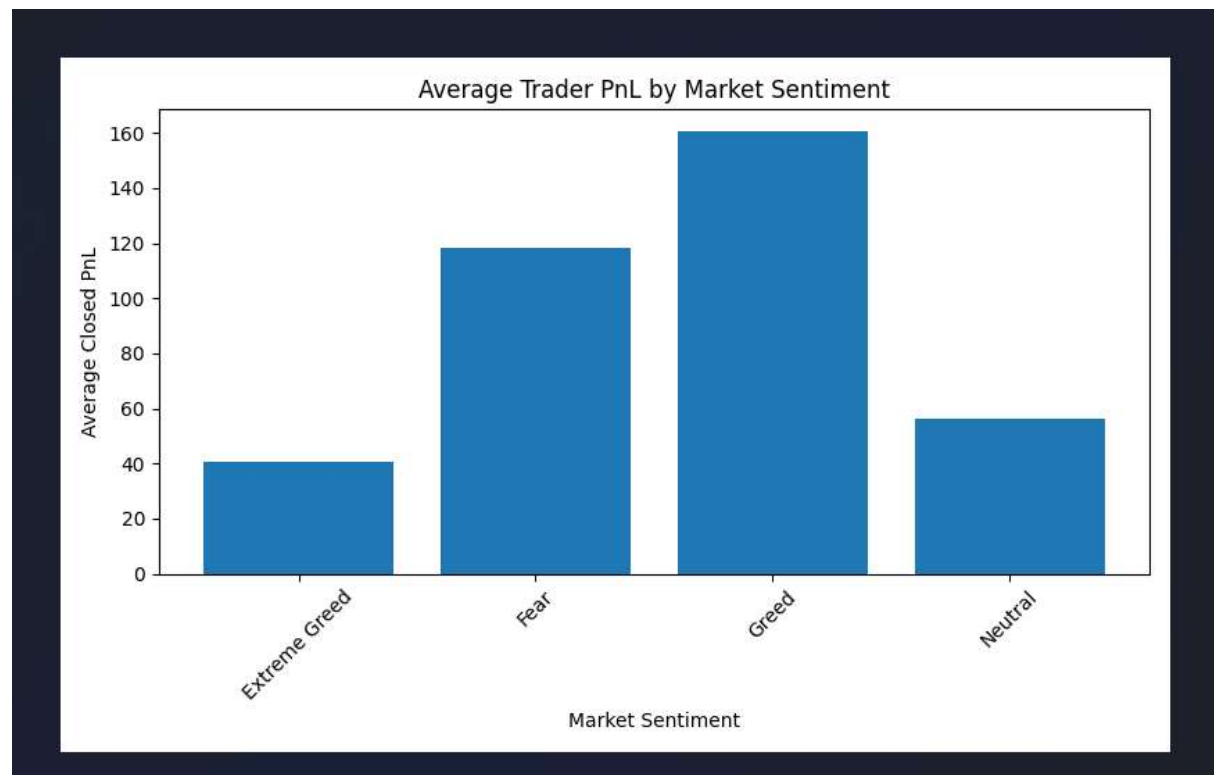


Figure 1: Average Trader PnL by Market Sentiment

7.2 Total PnL Analysis

Total PnL was calculated for each sentiment category to understand the overall profit or loss generated under different market conditions.

Insert Chart Here: Total PnL by Market Sentiment

Caption: Figure 2: Total Trader PnL by Market Sentiment

The highest total PnL was observed during Fear sentiment. This shows that although Greed had higher average PnL per trade, the overall profitability was stronger during Fear sentiment.

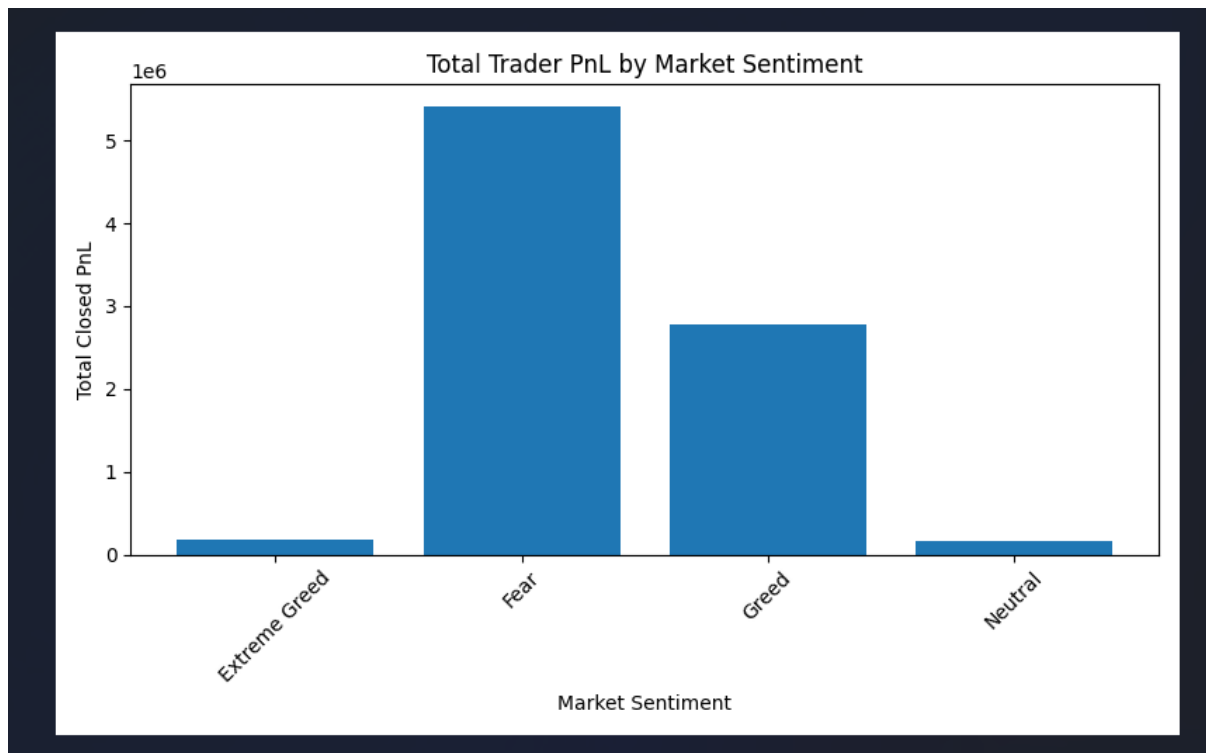


Figure 2: Total Trader PnL by Market Sentiment

7.3 Win Rate Analysis

Win rate was calculated by identifying profitable trades where Closed PnL was greater than zero. This helped measure how often traders made profit under different sentiment conditions.

Insert Chart Here: Win Rate by Market Sentiment

Caption: Figure 3: Win Rate by Market Sentiment

The highest win rate was observed during Fear sentiment. This means traders had a higher percentage of profitable trades during Fear market conditions.



Figure 3: Win Rate by Market Sentiment

7.4 Trade Side Analysis

Trade side analysis was performed to compare Buy and Sell performance under different sentiment conditions. This helped understand whether long or short positions performed better during Fear or Greed.

Insert Chart Here: Average PnL by Sentiment and Trade Side

Caption: Figure 4: Side-wise PnL Analysis

The analysis showed that trade side has an impact on profitability. Buy/Sell performance changed depending on the market sentiment.

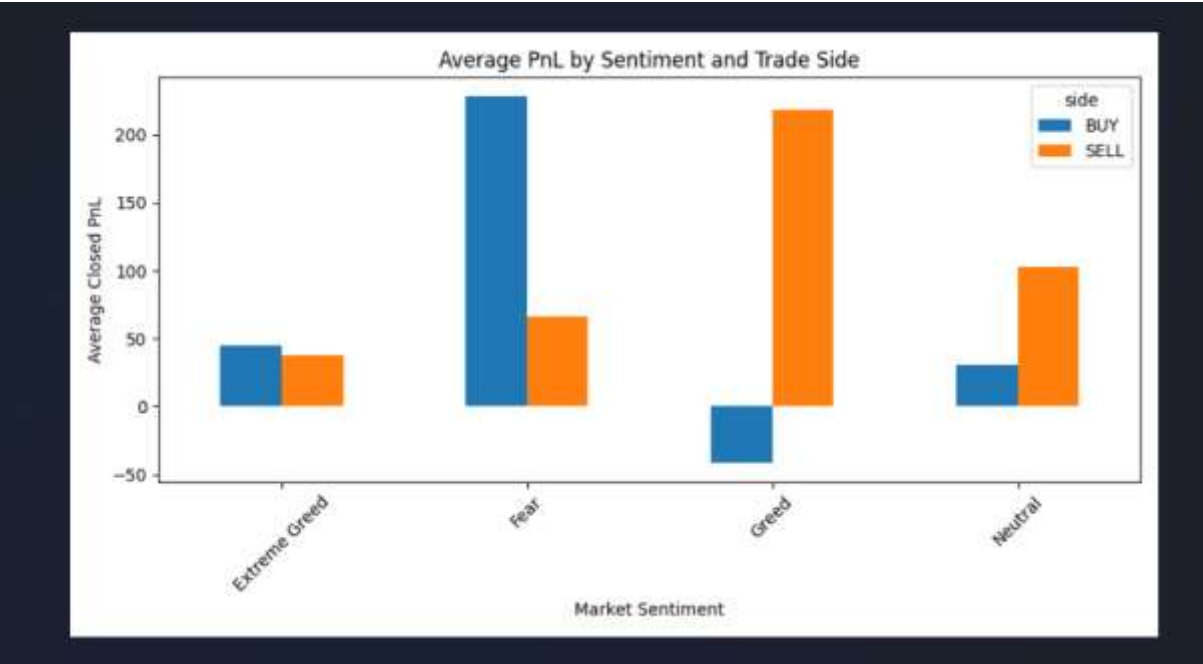


Figure 4: Average PnL by Sentiment and Trade Side

7.5 Symbol-wise Analysis

Symbol-wise analysis was performed to identify which trading symbols generated the highest total PnL.

Insert Chart Here: Top 10 Symbols by Total PnL

Caption: Figure 5: Top 10 Symbols by Total PnL

This analysis helped identify the most profitable symbols in the dataset. Some symbols performed better than others, showing that asset selection is important in trading strategy.

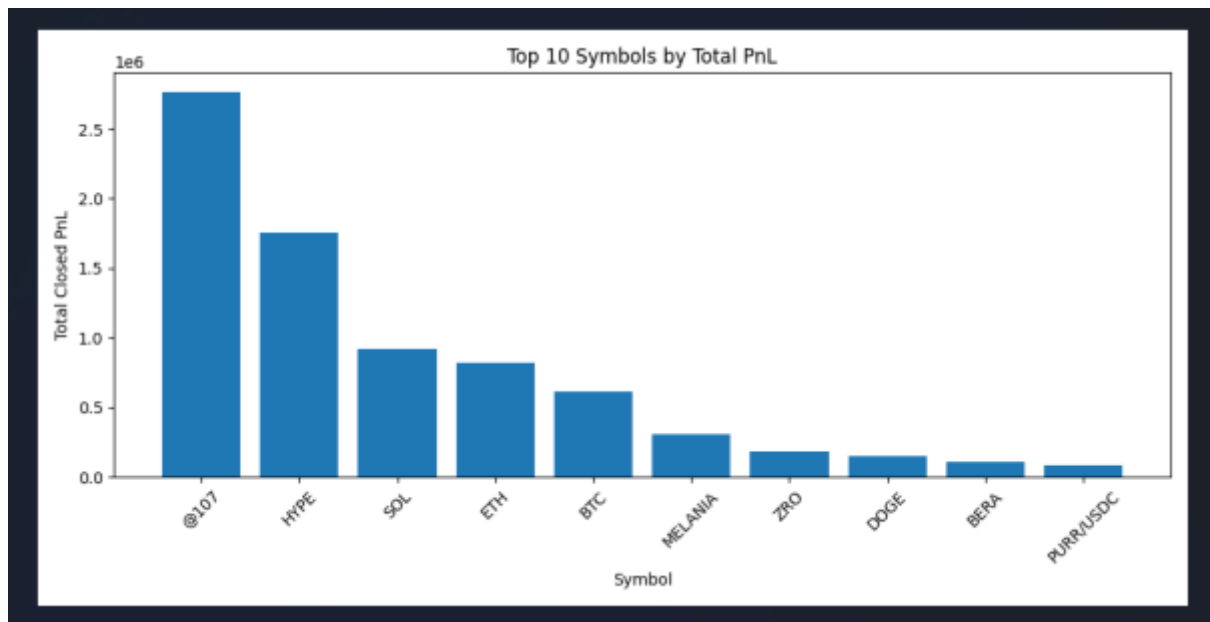


Figure 5: Top 10 Symbols by Total PnL

7.6 Account-wise Analysis

Account-wise analysis was performed to identify top-performing trader accounts. This helped understand whether some accounts consistently generated better profits than others.

The results showed that trader performance varies significantly across accounts. Some traders achieved higher total PnL and better average PnL compared to others.

7.7 Leverage Analysis

Leverage analysis was performed to understand how leverage affects trader performance and risk. Higher leverage can increase profit, but it can also increase losses.

The analysis showed that leverage usage varied across sentiment categories. This indicates that traders may change their risk-taking behavior depending on market conditions.

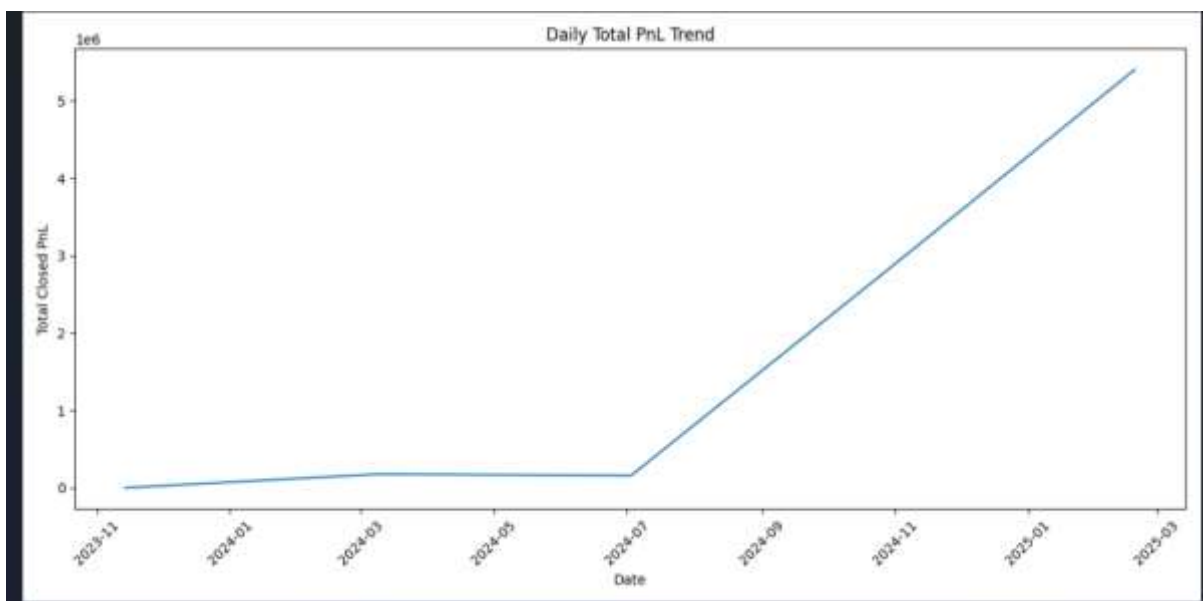
7.8 Daily PnL Trend

Daily PnL trend was analyzed to observe profit and loss movement over time.

Insert Chart Here: Daily Total PnL Trend

Caption: Figure 6: Daily Total PnL Trend

The daily trend helped identify periods of high profit and high loss. This shows that trader performance is not constant and changes over time based on market conditions.



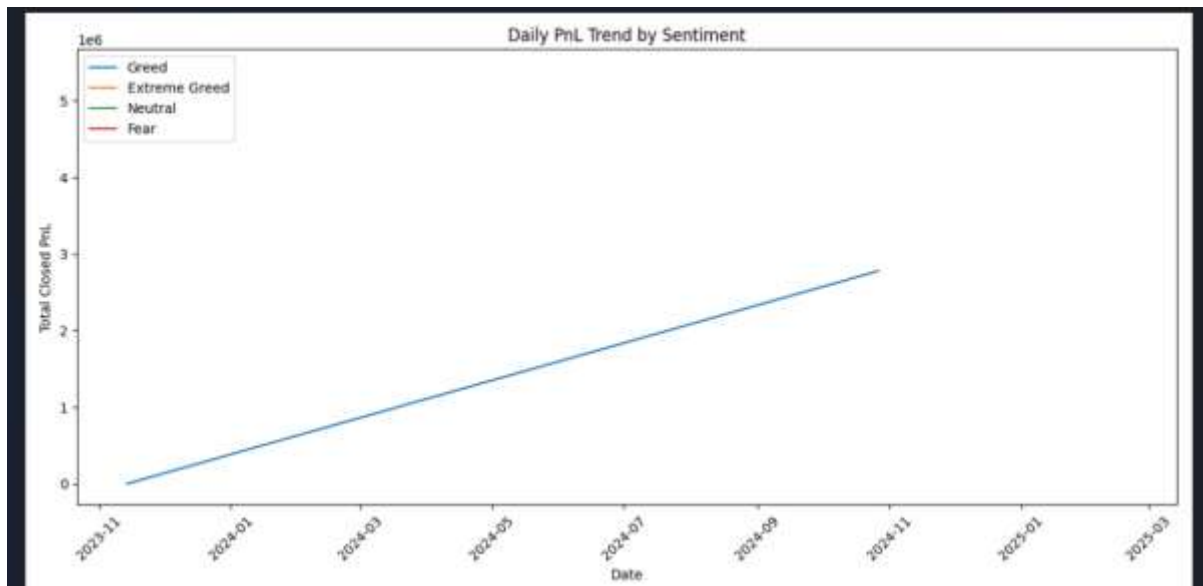


Figure 6: Daily Total PnL Trend

8. Key Findings

The key findings from the analysis are:

1. The highest average PnL was observed during Greed sentiment.
2. The highest total PnL was observed during Fear sentiment.
3. The highest win rate was observed during Fear sentiment.
4. Greed sentiment produced higher average profit per trade.
5. Fear sentiment showed stronger overall performance because both total PnL and win rate were higher.
6. Trade side, symbol, account behavior, and leverage influenced trader performance.
7. Leverage can increase both profit and loss risk.
8. Market sentiment has a clear impact on trading performance and trader behavior.

9. Business Insights

Based on the analysis, the following business insights can be derived:

1. Market sentiment can be used as an important factor in trading strategy.
2. Fear conditions may provide better overall trading performance because total PnL and win rate were higher.
3. Greed conditions may provide better average PnL per trade, but risk management is still important.
4. Traders should monitor leverage carefully because higher leverage can increase both profit and loss.

5. Symbol selection plays an important role in profitability.
 6. Account-level performance analysis can help identify consistent and high-performing traders.
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10. Conclusion

This project analyzed the relationship between Bitcoin market sentiment and trader performance. The analysis showed that trader profitability, win rate, trade side behavior, leverage usage, and symbol-wise performance change under different sentiment conditions.

The highest average PnL was observed during Greed sentiment, while the highest total PnL and highest win rate were observed during Fear sentiment. This indicates that Fear sentiment showed stronger overall performance, while Greed sentiment gave higher average profit per trade.

Overall, the results show that Bitcoin market sentiment has a clear impact on trader performance. These insights can help traders improve decision-making, manage risk, and build smarter trading strategies.

11. References

1. Bitcoin Market Sentiment Dataset
2. Historical Trader Data from Hyperliquid
3. Python Pandas Documentation
4. Matplotlib Documentation